

Homework 1

ENEE 132 — Basic Electronics I (Semiconductor Devices)
Due Date: 23 June 2026

Instructions:

Show all work clearly and express all final answers with correct units. Students must include their name and index number on all submitted solutions. Do not copy the questions. Provide the solutions only.

Question 1

- (a) Draw a clearly labelled diagram of a silicon *pn* junction at equilibrium, showing the p-type region, n-type region, depletion region, and barrier potential.
- (b) Explain the following within a *pn* junction
 - (i) Diffusion of majority charge carriers
 - (ii) depletion region formation and characteristics
 - (iii) barrier potential

Question 2

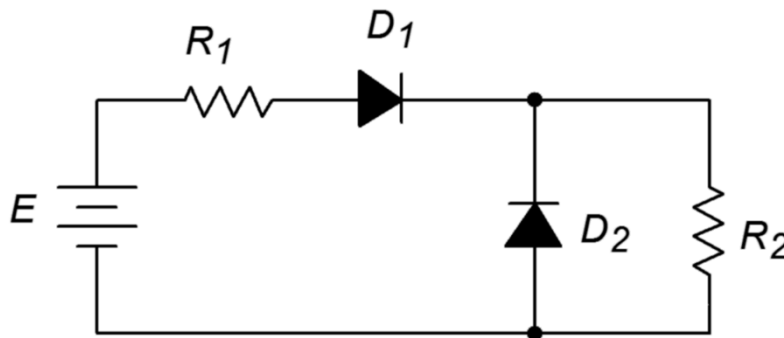


Figure 1

Refer to the circuit in **Figure 1**. The DC voltage source is $E = 15\text{ V}$ and the resistors are $R_1 = 8.2\text{ k}\Omega$ and $R_2 = 3.9\text{ k}\Omega$. Both D_1 and D_2 are silicon diodes.

- (a) State whether the diodes are forward-biased or reverse-biased
- (b) Determine the voltage and current for D_1 assuming the ideal diode model
- (c) Determine the voltage and current for D_1 using the practical diode model
- (d) Determine the voltage and current for D_1 using the complete diode model where $r'_d = 10\Omega$ and $r'_R = 100\text{ k}\Omega$.

*** End of Assignment ***